LESSON PLAN

PROPELLER THEORY (7)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	03
Total No. of Lesson Plans	8	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Co-axial and counter rotating propellers, forces acting on propeller
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Co-axial and counter rotating propellers, forces acting on propeller
5. Specific Objectives	7.1.1.1 Trainee will be able to describe the various types of propellers, principles of operation and pitch change mechanisms.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Demonstration of Co-axial propeller model; Safety: Handle model with care; Supervision: Instructor monitors trainee interaction
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What are the primary forces acting on a rotating propeller blade?
(b) How does a counter-rotating propeller system differ from a single propeller?
(c) Why are co-axial propellers used in certain helicopter configurations?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Definition of Co-axial Propellers • Advantages of Counter-Rotating Design • Torque Cancellation Principles • Efficiency Gains in Co-axial Systems • Applications in Helicopters and Drones	L & Ds	PPT, IB	15 min
(b)	• Aerodynamic Forces: Thrust and Drag • Centrifugal Force on Propeller Blades • Gyroscopic Precession Effects • Vortex Ring State in Co-axial Systems • Blade Element Momentum Theory Basics	L & Ds	PPT, PDF	20 min
(c)	• Structural Loads on Propeller Hubs • Vibration Damping in Co-axial Arrangements • Maintenance Considerations for Dual Props • Noise Reduction Benefits • Comparison with Pusher Propeller Configurations	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how counter-rotation eliminates torque reaction on the airframe.
(b) Describe the direction of centrifugal force relative to the propeller rotation axis.
(c) Identify two operational advantages of co-axial propellers over single propellers.

12. Home Assignment and Follow Up	Review LP 02 on Propeller Pitch Change mechanisms and prepare a brief note on how pitch variation affects the forces discussed in this lesson.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Constant Speed Unit (CSU)	Refer to the master syllabus reference list	LP 04

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Instructor Name, Instructor, P No. 12345	Senior Instructor, Senior Instructor, P No. 67890
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min